

API User Manual

Running the Flask App and Accessing Swagger UI

This repository contains a Flask application that provides APIs for accessing patient time series data. Follow the instructions below to run the application locally and interact with the APIs using Swagger UI.

OptiFox REST API

1.0.0

OAS3

api/v1/openapi.json

An API to fetch information for the PhysioNet DB and predict mortality and readmission rate

Servers

/api/v1

Patient

GET /patient/{stay_id} Fetch information of a patient

TimeSeriesFeature

GET /patient/{stay_id}/{features} Fetch time series data for a patient

OpenAPI 3.0.0 Format: The API documentation follows the OpenAPI 3.0.0 specification, providing a structured description of endpoints, parameters, and responses.

Interactive Documentation: Use Swagger UI to test various endpoints by providing required parameters and executing requests directly from the interface. Using the APIs

Prerequisites

- Python 3.10 or above
- Open the codebase in a dev container and build it using the [provided config](#) this should be fairly simple in vscode, you can also find further information on dev containers in the [dev containers tutorial](#)
- If you are somehow unable to accomplish the second prerequisite, in theory the APIs should also work in a virtual env and after installing all the requirements listed in the project [requirements](#)

After running the application in a dev container, run [server.py](#) to launch the flask server, then navigate to the below URLs depending on your interests.

API Documentation You can access the main flask API documentation in the below URL, this is not only a place where you are provided documentation on each API but also a playground to try out these APIs! <http://localhost:5000/api/v1/ui/>

Measure the performance 🚀 Although not sophisticated but we also provide prometheus metrics which can help understand th performance of the metrics, this requires more work but this basic setup lays the groundwork to further optimize the performance in the future. <http://localhost:3000/metrics>

Note: You cannot use the APIs without proper authorization

Authorization:

- First and foremost you have to login with the username "nani" and password "optifox" to generate the JWT token

Authentication

GET /auth/{user_id}/{password} Return JWT token

API to generate your JWT token

Parameters

Name	Description
user_id * required string (path)	User unique identifier
password * required string(\$password) (path)	Password associated with the user_id

nani

.....

Execute

Clear

2. After you generate the jwt token, from the response you need to login with the said token to be able to use the APIs

Available authorizations



jwt (http, Bearer)

Value:

Authorize

Close

Note: If you are using these APIs for your front end, make sure to send the token as part of the request header!

Endpoints: The Swagger UI will list all available endpoints along with their descriptions and input parameters.

Testing: Use the Swagger UI to test various endpoints by providing required parameters and executing requests directly from the interface.

Authentication	⌵
Patient information	⌵
TimeSeriesFeature	⌵
ReadmissionPredictionResult	⌵
Fetch patients in ICU with a timestamp	⌵

Versioning

1. /api/v1 does not require JWT tokens or any kind of authentication.
2. /api/v2 is the latest API version which does need token to access the APIs.

Logging There is also a logging capability where all the logs are written to a log folder which is autogenerated, this should give you further insights into the inner workings of the Flask APIs.